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Multicomponent magnetizations in the Foyers Old Red Sandstone (northern Scotland) and their bearing on lateral displacements along the Great Glen Fault

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SUMMARY

The Foyers Sandstone, a constituent of the Old Red Sandstone of northern Scotland, located on the eastern flank of the Great Glen Fault, has yielded a multicomponent palaeomagnetic structure. The principal magnetization (called the *B* component), with *in situ* mean direction *D*, $I = 185°$, $+8°$; $\alpha_{95} = 2.7°$, post-dates the local 'mid'-Devonian (Acadian) tectonic event, but this direction corresponds well with characteristic directions displayed by Devonian and Lower Carboniferous rocks of Scotland. The most significant overprinted component, the A2 magnetization, with *in situ* mean direction *D*, $I = 194°$, $-27°$; $\alpha_{95} = 4.1°$, has a more patchy occurrence than the *B* remanence, the strongest development of A2 being observed at sites that are situated close to the Great Glen Fault zone. The A2 direction is consistent with a Hercynian age. A third magnetization, the A1 component, of suggested Lower Tertiary age, has been found at one of the sites close to the fault zone. The A2 and *B* remanences support previously proposed declination discordances in corresponding magnetizations across the Great Glen Fault. It is concluded that the Foyers Sandstone developed its multicomponent magnetization through repeated tectonic events on the Great Glen Fault in Acadian, Hercynian and Alpine times respectively.

Key words: Foyers ORS, palaeomagnetism, Scotland, tectonics.

1 BACKGROUND

The Great Glen Fault, defining a prominent tectonotopographic boundary between the Northern and Southern Highlands of Scotland (Fig. 1), has long been accepted as an important zone of lateral displacement, but the question of sense and scale of movement has caused considerable debate (Kennedy 1946; Holgate 1969; Garson & Plant 1972; Winchester 1973; Storetvedt 1974, 1975, 1987; Mykura 1975; Donovan *et al.* 1976; Van der Voo & Scotese 1981; Storetvedt & Torsvik 1983; Smith & Watson 1983; Briden, Turnell & Watts 1984; Esang & Piper 1984; Hall 1986; Storetvedt & Otterå 1988; Rogers, Marshall & Astin 1989). In recent contributions to this discussion Storetvedt (1987) and Storetvedt and Otterå (1988) submit that the geological and palaeomagnetic evidence presently at hand is consistent with a two-phase megascale motion: a 600 km sinistral displacement in Acadian ('mid'-Devonian) time superseded by *circa* 300 km of dextral offset in the Hercynian. In fact, there is growing evidence that the whole of the British and Irish Caledonides may constitute a series of displaced blocks, bounded by the many prominent NE–SW striking faults that transect the British Isles, being amalgamated into their present juxtaposition by large-scale Palaeozoic transcurrent movements (i.e. Phillips, Stillman & Murphy 1976; Lambert & McKerrow 1978; Bluck 1984; Hutton 1987). The Dalradian of the Southern Highlands seems to have had a largely unconnected history relative to the Midland Valley block, illustrated, for example, by the fact that clasts of the Lower Old Red Sandstone (ORS) rocks south of the Highland Boundary Fault (HBF) show a provenance strikingly different from that of the Dalradian metamorphics presently occupying the adjacent terrain north of the HBF (Bluck 1984). However, geological contact across the HBF appears to have been established by the Upper ORS time, and evidence from the Firth of Clyde region suggests that the new palaeogeographic situation was linked to sinistral strike-slips along the HBF zone, probably

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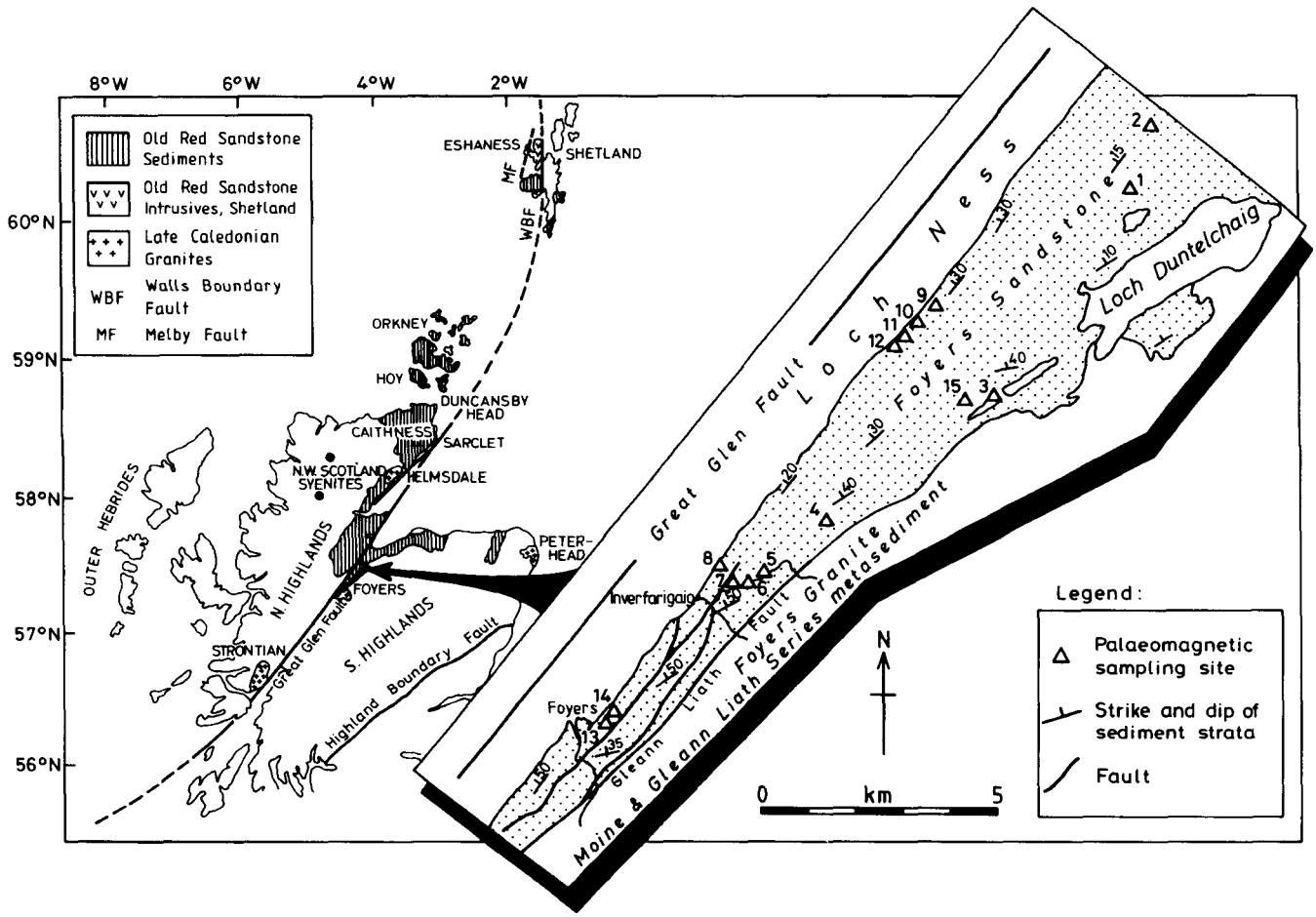


Figure 1. Sketch map showing the distribution of Old Red Sandstone rocks in northern Scotland. The Great Glen Fault, linking up with the Walls Boundary Fault through Shetland, provides a tectonic bound between the Northern and Southern Highlands. The geological insert map of the Foyers area, simplified after Mykura (1983), depicts the main tectonic features. The numbered triangles denote palaeomagnetic sampling locations.

of the order of hundreds of kilometres (Bluck 1984). This suggests that the fault zones along the Great Glen and the Highland Boundary may have experienced significant left-lateral shear in Acadian ('mid'-Devonian) time. The increasing evidence of displaced terranes in the British Caledonides suggests further that for palaeomagnetic evaluation of any strike–slip component along individual crustal dislocations only data from the two relevant tectonic blocks should be utilized; consideration of palaeomagnetic data from farther afield may easily lead to erroneous conclusions. It must be emphasized, however, that strike–slip movements across Scotland of the suggested scale (hundreds of kilometres) can only be detected palaeomagnetically through declination discordances, i.e. the fault zones in question must have an adequate curvature. This seems to be the case for the Great Glen Fault (see Flinn 1961, 1969), but probably not for other of the 'NE–SW' striking faults in Scotland.

In the case of the Great Glen Fault the palaeomagnetic database for the Southern Highlands block has been sparse (cf. Storetvedt 1987, and discussion below), and lack of reliable Devonian results in particular, for comparison with the now fairly substantial palaeomagnetic information from ORS rocks of Caithness and Orkney, has been a problem. The present study contributes needed information from ORS rocks east of the Great Glen Fault by providing detailed results from the Foyers Sandstone, Inverness-shire. Results from these ORS sediments were also reported in a previous palaeomagnetic study in the Foyers area (Kneen 1973), but these results have been questioned and reinterpreted by Morris (1976).

2 GEOLOGY AND PALAEOMAGNETIC SAMPLING

The Foyers Old Red Sandstone (Mould 1946; Stephenson 1972; Mykura 1983), a constituent of the Orcadian Basin of northern Scotland, is exposed in a 1–5 km wide belt along the SE shore of Loch Ness (Fig. 1). The sedimentary succession, in excess of 2 km thick, covers a varied suite of rocks, including breccias with locally derived granitic and metamorphic clasts, conglomerates, arkosic grits, fine-grained sandstones and subordinate shales (Mykura 1983). The floor of the Old Red Sandstone consists partly of Moine gneisses and granulites and partly of the Foyers Granitic Complex of presumed Silurian age (Miller & Brown 1965). The sedimentary succession, which is held to be Middle Devonian in age (Mykura 1983), has 'been deformed and

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Table 1. Palaeomagnetic collection details.

Sites*	Samples (FS)	Mean strike, dip	Site description
1	1-4, D63-69**	236, 13 NW	Poor natural outcrop
2	5-8, D70-76	225, 22 NW	Old quarry; yellow (? bleached) sandstone
3	9-12, D48-62	272, 38 N	Road Cut
4	13	194, 65 W	Natural outcrop
5	14-18	222, 45 NW	Natural outcrop in River Farigaig
6	19-22	232, 50 NW	- " -
7	23	252, 63 NNW	Poor natural outcrop
8	24-27	192, 40 W	Road cut
9	28-31	202, 25 W	- " -
10	32-35	213, 30 WNW	- " -
11	36-39	233, 28 NW	- " -
12	40-42	214, 27 WNW	- " -
13	43-46	275, 34 N	- " -
14	47	283, 35 N	- " -
15	D77-91	248, 33 NNW	Well-exposed natural outcrop

* See Fig. 1 for locations

** Samples denoted D are oriented drillcores. All other samples are oriented block samples

faulted by late-post-Middle Old Red Sandstone movements along the Great Glen Fault Zone' (Mykura 1983). However, the tectonic complexity is basically confined to the Foyers district in the south (see insert map, Fig. 1) where the sedimentary pile is cut and displaced by major and minor faults, with even local development of thrusting. North of Inverfarigaig the structural pattern is considerably simpler, the sedimentary layers displaying northwesterly dips varying between 10° and 50°.

The present study is based on two palaeomagnetic collections. The original one, comprising 47 oriented hand samples from 14 sites was later supplemented by 44 drillcores which provided more material from sites 1–3 as well as adding a new site, no. 15, to the collection. The sampled material consists principally of well-consolidated red siltstone (minor) and pink–red, medium–fine-grained sandstone. An exception is site 1 in which the rock is relatively soft, and irregular discolouring suggests that it has been subjected to weathering processes. Also the natural outcrops of sites 2, 4 and 7, the latter two only providing one oriented sample each, gave less fresh rocks than the bulk of the palaeomagnetic sites. According to the formational sub-divisions of Mykura (1983) the collected material comes from the Whitefield Formation (Fm.) (sites 1, 2 and 9–12), Creag a'Bhlair Fm. (site 8), Balchraggan Fm. (sites 3–7 and 15) and Foyers Fm. (sites 13 and 14). The Whitefield, Creag a'Bhlair and Balchraggan Formations are listed in descending stratigraphic order. The Foyers Fm. (southwest of Inverfarigaig) corresponds lithostratigraphically to the Whitefield Fm. (northeast of Inverfarigaig). The distribution of the palaeomagnetic locations is shown in Fig. 1, and further sampling details are listed in Table 1.

3 LABORATORY RESULTS

A total of 205 standard cylindrical specimens 1.9 cm in diameter were cut from the 91 samples and used in measurements. All were subjected to thermal demagnetization in steps up to 675°C. The bulk of the specimens was treated at Bergen University, employing a commercial (Schonsted TSM-1) and a laboratory-built demagnetizer; remanence was measured with Molspin magnetometer and, for weakly magnetic specimens, a Cryogenic Consultant magnetometer. A smaller collection of specimens was treated at Memorial University with a GSD-1 demagnetizer and a CTF cryogenic magnetometer. As the results from the two laboratories proved to be closely comparable, they have

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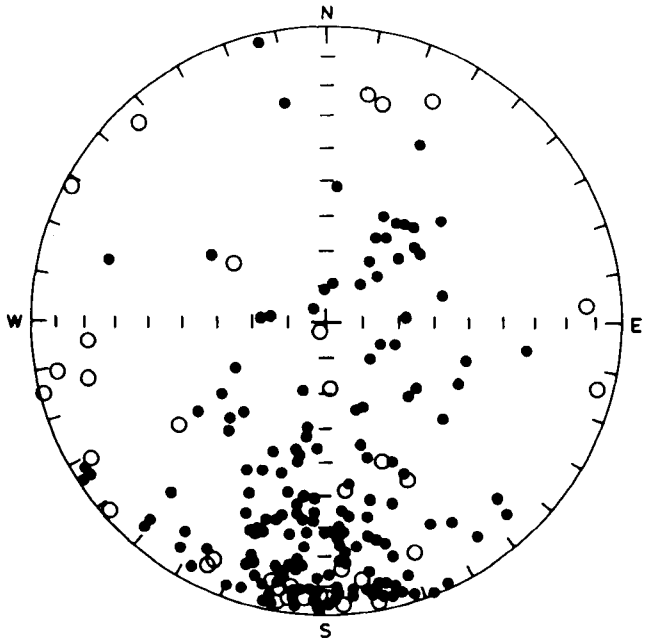


Figure 2. Directions of natural remanent magnetization (NRM) for all specimens that have been subjected to demagnetization studies. The diagram is equal area and closed (open) symbols represent downward (upward) inclinations.

been combined for the statistical treatment and will not be identified separately.

The directions of total natural remanent magnetizatrion (NRM) for all 205 specimens are depicted in Fig. 2. To a first approximation the overall directional pattern is suggestive of a reversed magnetization having low inclination, plus at least one subordinate component of normal polarity. As we show below the thermal analysis has indeed confirmed that the shallow reversed magnetization is the dominant characteristic component of the Foyers Sandstone. In the palaeomagnetic terminology of the Orcadian Basin this flatlying magnetic vector has been termed the *B* magnetization (Storetvedt *et al.* 1978; Storetvedt & Carmichael 1979). Wherever it has been observed (including the Foyers Sandstone), this magnetization shows a dual-polarity structure, but the reverse component provides the predominant contribution. The superimposition of normal and reverse *B* remanences may define resultant NRM directions in any intermediate position. Fig. 3 gives a number of examples of component interplay as revealed by thermal demagnetization.

Specimen FS20-C3 (Fig. 3a) describes the most common vectorial pattern encountered in the present study. After removal of a minor soft and steeply inclined magnetization, associated with a slight southward progression of the resultant vector, the trend reverses and the successive vectors define a northerly trending great circle path during which a shallow southerly, generally downward pointing, *B* magnetization is erased. Practically all northerly swinging remagnetization paths are disrupted by experimental noise at various temperatures above *circa* 550 °C, obliterating resolution of the high-temperature normal component; therefore the majority of specimens carrying the *B* component show vector trajectories that by-pass the origin (the erratic stage is disregarded and not shown in the diagrams). When extended, the northward trending remagnetization circles intersect in a shallow position almost due north, corresponding to a normal *B* component direction, but only two specimens, FS6–A1 and FS44–A1 (Fig. 3b, c and e), seem to define this normal component unequivocally. The latter sample is interesting in that it has an unusually strong normal *B* magnetization, but switches into the reverse *B* direction at the final step of demagnetization (650 °C). Among the analysed specimens that do not contribute to the final palaeomagnetic database, the majority can be grouped in one of the following two categories: (a) specimens that define curved vector plots (such as FS6–A1, below 550 °C), and (b) specimens that have a 'banana'-shaped remagnetization trend (i.e. FS12–B1 and FS25–A1). The latter category of specimens most likely carry more than two superimposed magnetizations with interleaved thermal spectra. While some of these specimens do seem to display an 'acceptable' *B* magnetization in vector plots their acceptance for palaeomagnetic directions could have easily led to artificiality in component extraction.

All palaeomagnetic directions deduced from the present study are based on linear patterns in vector diagrams. The overall quality of the data can be judged from several examples presented in Figs 3–6. Results from the four sampling locations (1, 2, 4 and 7) considered least promising from field observations, proved to be indeed too complex and scattered to be useful for analysis. However, the remaining 11 sites give consistent palaeomagnetic directions and disclose a more clear-cut and informative database than was the case in previously reported individual Orcadian Basin studies. The *B* magnetization is present throughout, but additional components, A1 and A2, contribute significantly to the NRM at some sites. The A2 component, with southerly declinations and an average upward inclination of *circa* 25 ° has been found at seven sites (nos 3, 5, 8, 11, 12, 13 and 14). When co-existing with the reverse *B* magnetization, the A2 remanence occupies with one exception (Fig. 6a) the higher of the two blocking temperature ranges (cf. Fig. 5c and d). It is frequently seen, however, that at the highest steps of demagnetization the total remanence vector moves into normal polarity suggesting that the residual component is the normal *B* magnetization. This suggests that the A2 component is interleaved between the two antiparallel *B* vectors, implying that for successful separation of the A2 magnetization a sufficiently broad window between the normal and reverse *B* component spectra is required. This condition may not always be present, suggesting that the A2 remanence could be a more important palaeomagnetic constituent in this study than is actually reflected by the number of resolved directions (26, see Table 2 and Fig. 7).

Site 12 (Fig. 6) differs from all other sites by having a significant low blocking temperature ($T_b \leq 100$ °C) magnetization. The remanence erased at this initial stage of demagnetization, the A1 component, was only uncovered at this site and is directed nearly due north, with intermediate downward inclinations. Site 12 provides the full range of palaeomagnetic information, displaying a variety of component combinations. For example, the A2 and *B* remanences may overlap to such an extent that only the A1 magnetization is successfully disclosed, or one can find the

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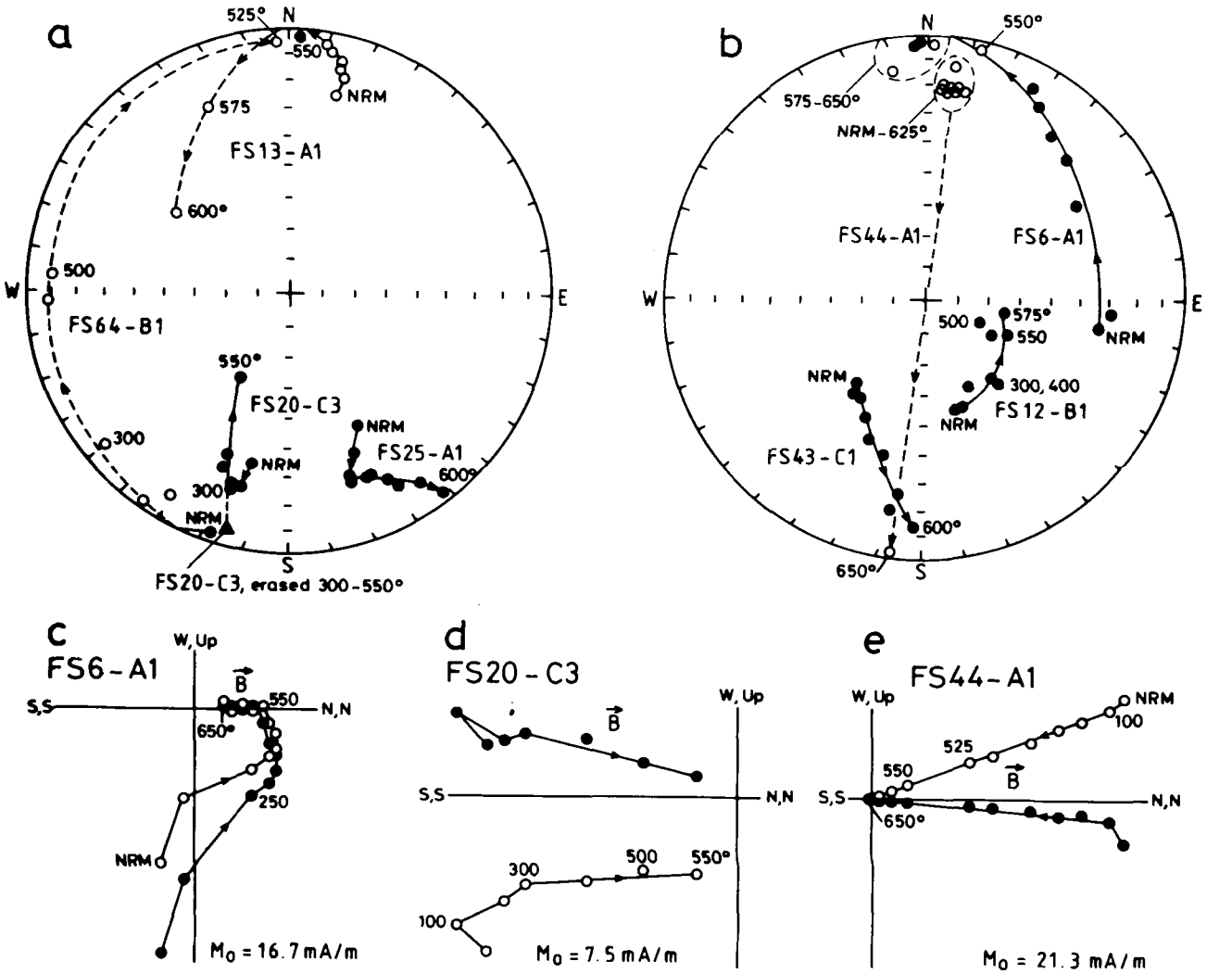


Figure 3. Examples of different types of multicomponent interplay encountered in the Foyers Old Red Sandstone (cf. text for discussion). Diagrams (a) and (b) are equal area plots. Diagrams (c–e) are orthogonal vector projections of three specimens from diagrams (a) and (b). In the vector plots solid (open) symbols are points in the horizontal (vertical) plane. B refers to the *B* component. Other conventions as for Fig. 2.

A1 and *B* components in co-existence (Fig. 6e). A number of examples (Fig. 6b–d, f) show the A1 and A2 components in total dominance, having suppressed the *B* remanence beyond recognition. Finally one specimen, FS40–A, shows all three magnetizations together (Fig. 6a). Fig. 7 shows the distribution of the three Foyers ORS magnetizations at specimen level.

In view of the relatively strong tectonic deformation in the southern part of the sedimentary complex, and as a precaution against unrecognized anisotropy effects caused by any superimposed structural fabric, measurements of anisotropy of magnetic susceptibility (AMS) were carried out on five specimens from each of the 10 principal palaeomagnetic sites. The measurements show relatively low anisotropies, varying between 1.8 and 9 per cent (average 4.3), and the AMS data give mostly well-defined oblate magnetic ellipsoids (Fig. 8). With two exceptions, sites 3 and 15, the magnetic foliation plane is oriented very close to the measured bedding plane, suggesting that the AMS results are principally governed by a primary bedding-oriented fabric. The maximum susceptibility (K_{max}) varies both within and between the sites, but the two directional populations that dominate the Foyers ORS, components *B* and A2, show consistent groupings (see also Fig. 9 for *B*-component site mean distribution). Hence, we cannot infer any noticeable relationship between the palaeomagnetic directions and the AMS structure.

4 INTERPRETATION; TECTONOMAGNETIC ASPECTS

As outlined above the Foyers Sandstone has a multicomponent palaeomagnetic structure, but splitting the magnetization into its components has been effectively accomplished by thermal demagnetization. The directional populations of the individual remanences are well grouped (Fig. 7), the overall *in situ* magnetization data for each group being listed in Table 2.

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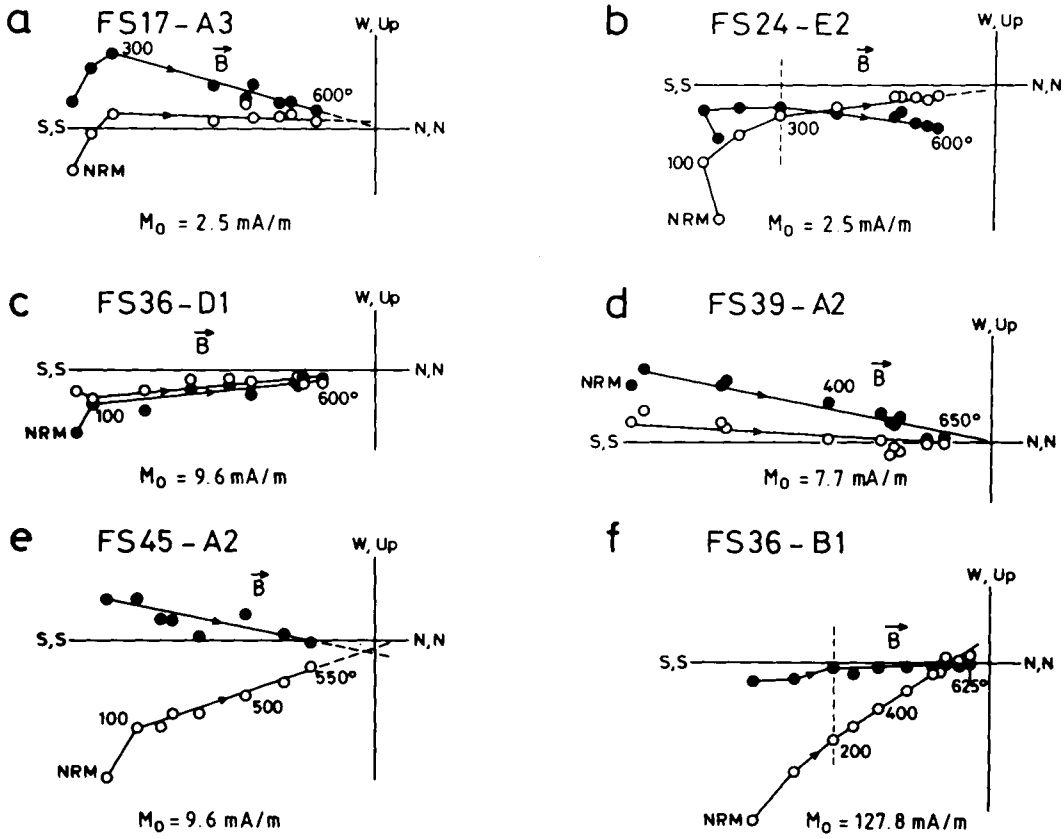


Figure 4. Vectorial displays of examples of the predominant magnetization of the Foyers Sandstone, the *B* component. Diagram conventions are as for Fig. 3. M_0 is the intensity of NRM.

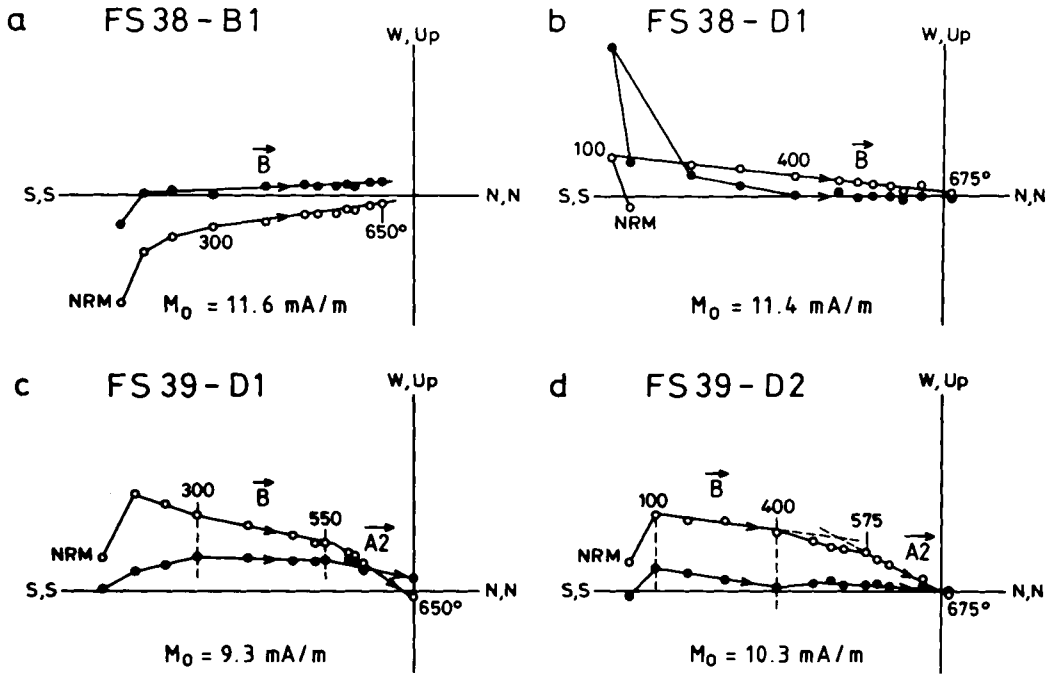


Figure 5. Further demagnetization results for site 11, in additon to those of specimens FS36–B1 and D1 and FS39–A2, in Fig. 4. Site 11 is a typical *B*-component location, but the A2 magnetization is also occasionally found (cf. c and d). Diagram conventions are as for Fig. 3.

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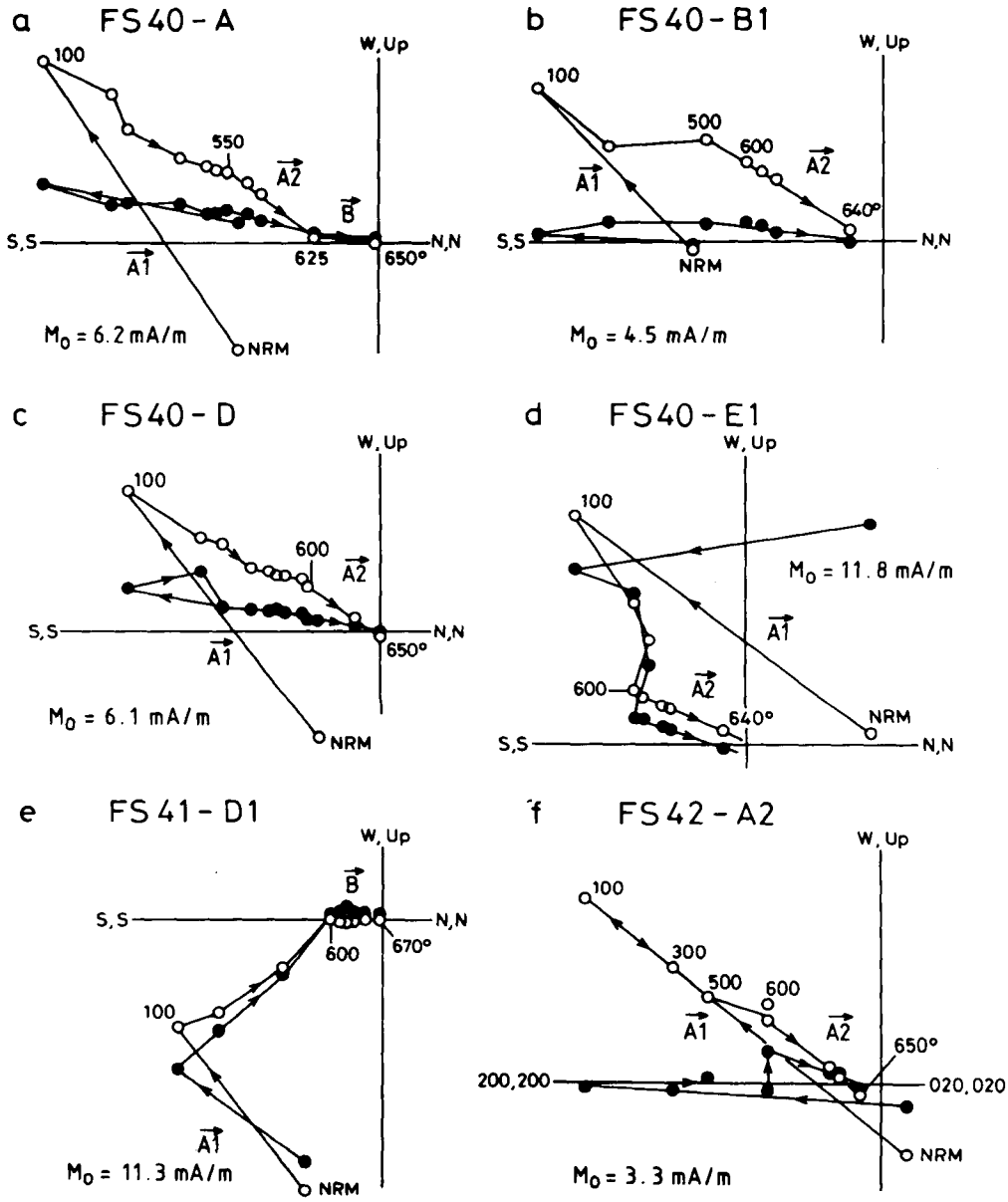


Figure 6. Examples of the magnetization structure of site 12 which is dominated by secondary overprints, components A1 and A2. The *B* remanence is only seen sporadically (cf. a and e). Specimen FS40–A (a) seems to possess all three magnetizations. Diagram conventions are as for Fig. 3.

Table 2. Overall palaeomagnetic data for the Foyers Old Red Sandstone.

Comp.	Suggested magn. age.	N	D_m	I_m	α_{95}	K	Pole
A1	M/T	18	017.6	47.7	4.3	64.9	$144.4^E, 50^N$
A2	P/C	26	193.6	-27.4	4.2	47.1	$156.3^E, 46{\cdot}1N$
B	D_{m-u}	81	184.8	8.1	2.7	35.6	$170.2^E, 28{\cdot}6N$
B^a	- " -	11	185.8	6.7	4.4	107.5	$168.8^E, 29{\cdot}2N$

M, T, P, C and D denote Mesozoic, Tertiary, Permian, Carboniferous and Devonian respectively. Other symbols are: m, middle; u, upper; N, number of unit vectors (i.e. specimens, if not otherwise stated); D_m, mean declination; I_m, mean inclination; α_{95}, radius in the 95% circle of confidence; K, precision parameter (Fisher, 1953). The B^a-data are based on unit weight on site mean directions.

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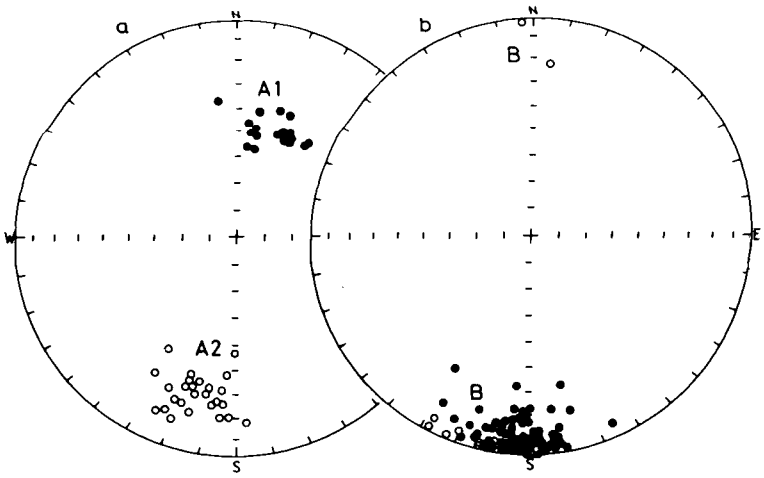


Figure 7. Palaeomagnetic results at specimen level. Diagram (a) shows overprinted magnetizations while (b) depicts the distribution of the *B* remanence. Overall palaeomagnetic data are given in Table 2. Projections are equal area and conventions are as for Fig. 2.

4.1 *B* component

The *B* component, which is the predominant magnetization in these sediments, has a mean direction $D/I = 185°/8°$, $\alpha_{95} = 2.7°$. As is seen from Fig. 9, the stepwise structural correction of the sandstone data leads to progressive increase in dispersion. Hence, the magnetization is inferred to be post-tectonic, so that the *in situ* directions are those of relevance for further palaeomagnetic consideration. The negative fold test is not surprising as both the structural and palaeomagnetic histories of rock formations adjacent to the Great Glen Fault are prone to be strongly linked to phases of tectonic reactivation of the fault zone (Storetvedt & Meland 1985; Storetvedt 1987). Mykura (1983) has

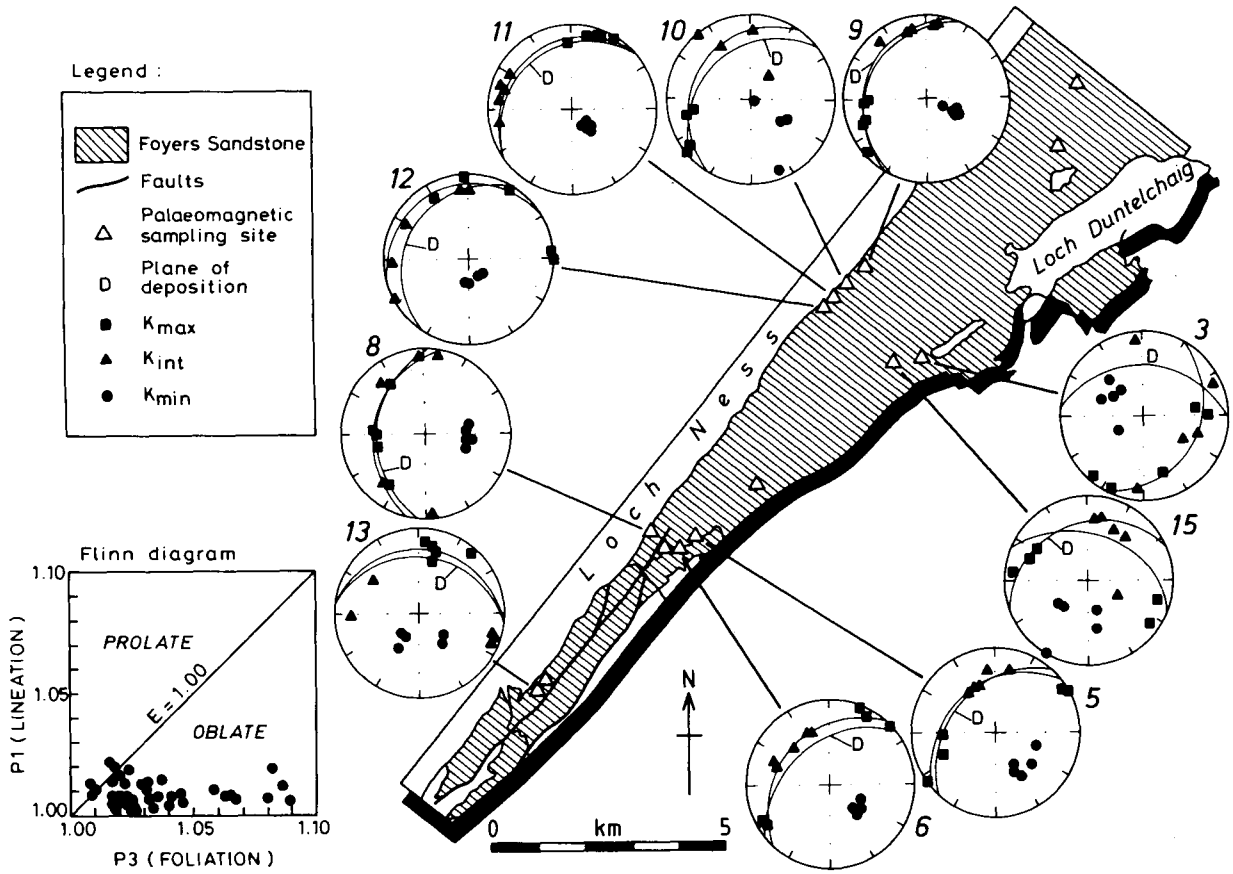


Figure 8. Magnetic fabric for specimens from the main palaeomagnetic locations. In general, the results define oblate susceptibility ellipsoids (see Flinn diagram), and the resulting magnetic foliation planes, shown by downward dipping great circles, agree closely with the bedding plane, D.

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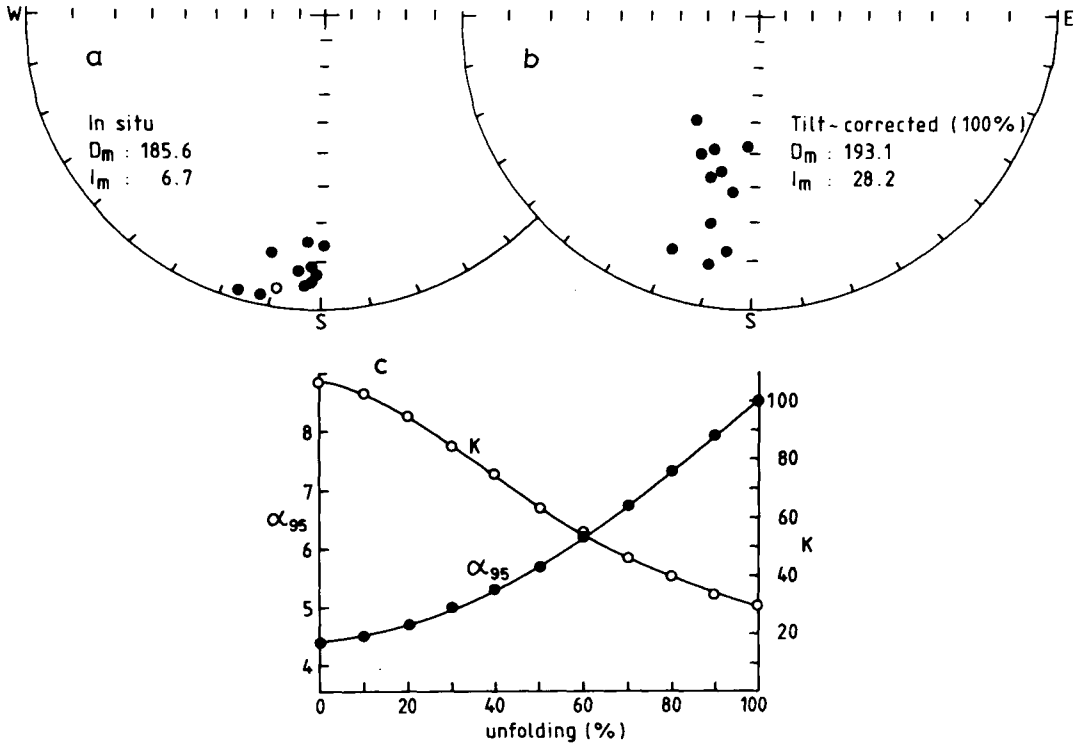


Figure 9. Unfolding test of the Foyers Old Red Sandstone *B* magnetization based on site mean directions D_m, I_m, K and α_{95} are the declination, inclination, precision parameter and the radius of the 95 per cent confidence circle of the overall mean vector ($N = 11$ sites). The stepwise structural correction (c) leads to a progressive increase in directional dispersion. Diagrams (a and b) are stereographic and conventions are as for Fig. 2.

suggested that the tectonic deformation of the Foyers Sandstone was the result of movements along the Great Glen Fault in the late-post-Middle Devonian. Tectonic movements would no doubt produce an enhanced thermochemical environment capable of causing both remagnetization and diagenetic processes in the Foyers Sandstone, the effects continuing an unknown length of time after the actual tectonic event had ceased. As we argue below the *in situ*, i.e. post-tectonic, magnetization is consistent with a Middle–Upper Devonian age but it seems to predate the Acadian age strike–slip movement on the Great Glen Fault associated with the observed Siluro-Devonian palaeomagnetic discordances across the fault (see below).

4.2 A2 component

The overall direction of the A2 component, $D/I = 194°/-27°$, $\alpha_{95} = 4.2°$, gives a palaeomagnetic pole (Table 2) that falls within the well-defined Upper Carboniferous–Permian polar group for Europe. By Permo-Carboniferous time the Foyers Sandstone is likely to have been in a well-consolidated state, imposed by the Acadian tectonic event, with reduced vulnerability for acquiring magnetic imprints. This may account for the observations that the A2 magnetization is more weakly developed and has a more restrictive areal coverage than the *B* component, being principally found in sites along the shore of Loch Ness, i.e. in the closest possible position to the Great Glen Fault where the remagnetization environment (caused by tectonic reactivation) is assumed to have been most intense.

4.3 A1 component

The A1 magnetization has only been found at one site, again located along the shore of Loch Ness, at which it carries a significant magnetic moment with $T_b \leq 100\,°C$. In previous studies of Orcadian Basin rocks magnetizations equivalent to the A1 component have been found only in the poorly consolidated (i.e. porous) John O'Groats Sandstone (Storetvedt & Carmichael 1979) and the magnetically soft Duncansby Neck (Storetvedt *et al.* 1978). Both these rock units are located fairly close to the Great Glen Fault in NE Caithness (Fig. 1). In line with the foregoing arguments only a modest tectonomagnetic process, confined to formations situated close to the fault, need be invoked to explain the spot-like occurrences of the A1 remanence which are associated only with rocks that are particularly susceptible to thermochemical remagnetization. In previous studies this latter remanence was loosely referred to as Mesozoic in age, but recent consideration of the Meso-Cainozoic palaeomagnetic database for Europe (Storetvedt 1990) would favour a Lower Tertiary age for the A1 magnetization. Marine geophysical studies suggest that shear motions on the Great Glen Fault in Alpine time were only of the order of tens of kilometres (Flinn 1975; McQuillin, Donato & Tulstrup 1982), i.e. much less than can be inferred for Acadian and Hercynian times (Storetvedt 1987). These results give substance to the view that the degree of development of individual magnetization components in rock formations adjacent to the Great Glen Fault is closely tied to the intensity of the related tectonic movements.

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4.4 General tectonomagnetic implications

In the Foyers area, Marston (1971) observed that the strong faulting and cataclasis affecting the basement of the Foyers Sandstone were in marked contrast to the much more modestly developed local structural deformation seen in the Old Red Sandstone itself: He argued therefore that strong faulting and cataclasis along the Great Glen Fault occurred in pre-Old Red Sandstone time, later reactivations of the fault zone having been on a smaller scale. The Foyers granite and other similar Caledonian intrusions along the fault do most likely originate from a major Silurian tectonic event. However, the location within the fault zone of these granite bodies would make them prone to later tectonic remagnetization (at various stages of fault reactivation). One should expect therefore that the Foyers, Strontian and Helmsdale granites would possess palaeomagnetic structures similar to that of the Foyers Old Red Sandstone. However, the presently available results from these intrusive bodies (Torsvik, Løvlie & Storetvedt 1983; Torsvik 1984), which were considered adequate for dismissing a previous proposal of an extravagant relative motion of the Northern Highlands (Van der Voo & Scotese 1981), lack the precision now required for a reasonable palaeomagnetic comparison astride the fault. For example, the published overall characteristic directions of magnetization of the Helmsdale and Strontian granites, both being located on the west flank of the Great Glen Fault, are significantly different at the 95 per cent probability level. It is most likely that this directional discrepancy reflects the effect of unresolved multicomponent remanences (Torsvik 1984). Both the Helmsdale granite and the Sarclet Old Red Sandstone (see Fig. 1) display *B* magnetizations that are smeared in the direction of declination, and such results must therefore be disregarded when discussing declination discordances across the Great Glen Fault of the order of a few tens of degrees (cf. Storetvedt 1987). The Strontian granite shows evidence of the A1 component, but its main magnetization is ill-defined, apparently lacking the necessary detailed experimental and analytical treatment. Finally, at first sight the overall remanence direction of the Foyers granite, $D/I = 188°/9°$ (Torsvik 1984), seems indistinguishable from that of the *B* magnetization of the Foyers Old Red Sandstone. However, a closer inspection of the granite data indicates that the overall direction may be artificial, constituting the sum of two remanence populations: one with shallow downward ('*B*') and one with shallow upward (A2) inclinations. Further work is clearly needed to clarify the magnetization structure of the Foyers granite which most likely possesses superimposed palaeomagnetic components similar to that unfolded for the adjacent Old Red Sandstone.

Recent studies (Trench *et al.* 1989; Torsvik *et al.* 1989) have claimed that the widely distributed shallow *B* magnetization is of post-Devonian age, and that the true relative Devonian field had intermediately steep inclinations as alledgedly recorded by the Lower Devonian lavas of Scotland. However, reinvestigation of one of these lava sequences, the Cheviot Complex, Storetvedt *et al.* (in preparation) have uncovered that the magnetic mineralogy is in a state of extensive low-temperature oxidation and that the previously reported palaeomagnetic results are basically

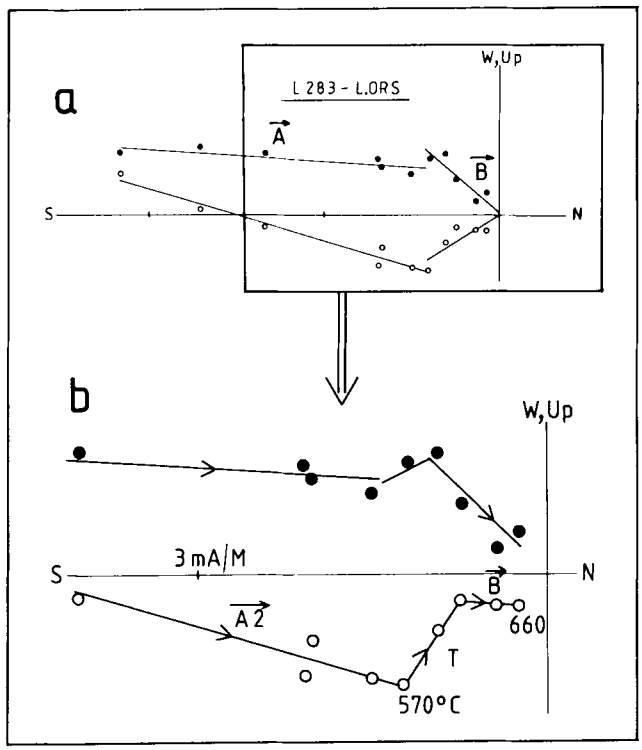


Figure 10. Orthogonal vector projections of a thermally demagnetized Lower Old Red Sandstone sample from the Midland Valley, according to data by Torsvik *et al.* (1989). Diagram (b) takes a closer look at the high-temperature results, arriving at a component interpretation different from that of the original authors (a). See text for detailed discussion. Diagram convention as for Fig. 3.

unresolved multicomponent magnetizations. Again, the underlying single-component palaeomagnetism concur with the widespread shallow *B* axis.

A recent example from the Devonian of the Midland Valley (Torsvik *et al.* 1989) may serve as a proper demonstration of analytical mistakes frequently seen even in the most modern palaeomagnetic work from Scotland. Thus, Fig. 10(a) shows the vector plots of a specimen from Torsvik *et al.* (1989), the only example they give from their Devonian sediments. As can be seen from Fig. 10(a) they delineate a high-temperature component with SW declination and intermediately steep downward inclination. This latter direction, however, is not consistent with their data. Fig. 10(b), which is a blow-up of part of their diagram, shows clearly that the high-temperature (haematite) magnetization is practically horizontal. The horizontal vector plot is ill-defined at temperatures above 570 °C, but it seems that this specimen has in fact superimposed A2 and *B* remanences, precisely as we have seen above in many samples of the Foyers Old Red Sandstone. The *T* trajectory of the vertical plot of Fig. 10(b) has no corresponding linear branch in the horizontal vector plot suggesting that the temperature range concerned represents the partial spectra overlap of the A2 component and the sub-horizontal *B* magnetization.

The incomplete and superficial remanence analyses seen in many studies of Scottish Palaeozoic rocks confuse geophysical and geological issues. Therefore, any 'realistic' palaeomagnetic comparison across the Great Glen Fault,

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from Southern to Northern Highlands, must presently be based on a selected database. However, such a selection process is admittedly difficult as most studies do not provide the necessary experimental/analytical documentation that is required for assessment of component reliability in multiphase palaeomagnetism. The results from northern Scotland used in the present palaeomagnetic comparison are listed in Table 3, and the individual mean directions are plotted in Fig. 11.

The Foyers Sandstone *B* component provides further substance to previous claims (Storetvedt 1974, 1987) in terms of a declination discrepancy in Old Red Sandstone data across the Great Glen Fault (direction 5 versus 7–11). Across the Melby Fault in Shetland (Fig. 1) there seems to be another declination discordance in the same sense (direction 7–11 versus 12). The pattern of assumed Silurian age magnetizations add further to the notion of a westward shift in palaeomagnetic declination from Southern to Northern Highlands (direction 1 versus 2–4). The westward swings of the palaeomagnetic vectors of the Northern Highlands, relative to the Southern Highlands, can be associated with net left-lateral shear motion along the Great Glen Fault and the Melby Fault. The Hercynian overprints for rocks in the Northern Highlands block define an overall easterly spread of their mean directions relative to the 'reference' magnetization of corresponding age in the

Table 3. Palaeomagnetic data from Northern and Southern Highlands of Scotland referred to in Fig. 11.

No	Formation	Magnetic age	N	D_m	I_m	95	Pole	Reference	Remarks
	SOUTHERN HIGHLANDS								
1	Peterhead Granite	S	38	180	22	5.2	178^E, 21^N	Torsvik (1985)	
5	Foyers ORS B-comp.	D_{m-u}	81	185	8	2.7	170^E, 29^N	this paper	
16	Foyers ORS A2-comp.	P/C	26	194	-27	4.2	156^E, 46^N	this paper	
17	Peterhead dyke	P/C	64	193	-19	1.3	162^E, 41^N	Torsvik (1985)	
	NORTHERN HIGHLANDS								
4	Borrolan inner phase	S	37	032	-14	6.3	141^E, 20^N	Turnell & Briden (1983)	
3	Loch Ailsh	S	28	033	-22	7.6	141^E, 15^N	- " -	
2	Assynt Dykes, combined	S	25	027	-24	5.2	141^E, 16^N	- " -	
6	Orkney dykes	D_{m-u}	32	169	4	3.1	190^E, 28^N	Storetvedt & Otterå (1988)	
7	Lower Caithness ORS	D_m	27	205	3	6.4	149^E, 27^N	Storetvedt & Torsvik (1983)	
11	John O'Groats ORS	D_m	12	209	6	10.4	145^E, 24^N	Storetvedt & Carmichael (1979)	NRM dominated by post-Palaeozoic overprint
8	Duncansby neck	D_{m-u}	20	205	8	6.3	149^E, 24^N	Storetvedt et al. (1978)	- " -
9	Orkney volc.	D_{m-u}	7	205	8	9.1	150^E, 24^N	Storetvedt & Petersen (1972)	
10	Hoy ORS rocks	D_{m-u}	17	208	9	5.5	146^E, 23^N	Storetvedt & Meland (1985)	
12	Esha Ness ignimbrite	D_m	55	221	2	2.5	134^E, 21^N	Storetvedt & Torsvik (1985)	
15	Helmsdale, A-comp.	P/C	21	183	-31	6.0	172^E, 49^N	Torsvik et el. (1983)	
14	Hoy ORS, A-comp.	P/C	7	176	-31	4.2	182^E, 48^N	Storetvedt & Meland (1985)	
13	Orkney dykes, A-comp.	P/C	42	172	-24	3.3	189^E, 43^N	Storetvedt & Otterå (1988)	
18	Vol. and sed. rocks, SE. Orkney	P/C	-	191	-20	5.9	163^E, 41^N	Robinson (1985)	

S, D, C and P denote Silurian, Devonian, Carboniferous and Permian respectively.
Other symbols are as for Table 2.

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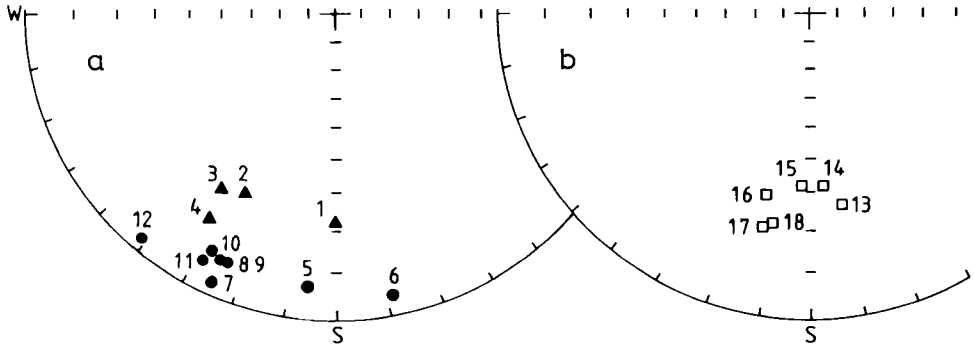


Figure 11. Distribution of suggested (a) Silurian (triangles), Devonian (circles) and Permo-Carboniferous (squares) magnetizations of north Scotland. Only assumed reliable data from the Northern and Southern Highlands are considered (see text and Fig. 1). The numbers refer to mean remanence directions for the rock formations listed in Table 3. Directions 2–4 are shown with inverted polarity for easier comparison with the other results. Projection and conventions are as for Fig. 2.

Southern Highlands (cf. Fig. 11b). This latter pattern is consistent with right-lateral shear along the Great Glen Fault (Storetvedt 1987; Storetvedt & Otterå 1988) in Permo-Carboniferous (Hercynian) time, impressing secondary magnetizations (A2 components) at various stages of the relative tectonic rotation. Also, the left-lateral shear, which is inferred to be of Acadian (Middle–Upper Devonian) age added syn-post-tectonic *B* remanences to the pre-tectonic *B* magnetizations in some rock formations, providing total *B* populations with declination smear (cf. discussion in Storetvedt 1987). Another effect of shear motions would be dyke intrusion. The anomalous easterly declination of the *B* magnetization of the Orkney dykes (direction 6, Fig. 11a) can therefore be readily accounted for (see Storetvedt & Otterå 1988).

Due to the distances across Scotland being small, and the fact that the geological boundaries of the Southern and Northern Highlands (separating the Lewisian basement and the Moinian and Dalradian metasediments) and the regional metamorphic trends run closely parallel to the Great Glen Fault zone (Winchester 1973; Storetvedt 1974, 1987), it is very difficult to set reasonable geological constraints on the scale of transcurrent motion. However, the closely similar inclination of comparable magnetizations across the fault (cf. Fig. 10) suggest strongly that mid–late Palaeozoic relative latitudinal displacements, between what are now the Northern and Southern Highlands, could not have exceeded a few degrees. In fact, it is the curvature of the fault, providing relative block rotations and associated declination discordances, that has rendered palaeomagnetism an effective tool for studying the transcurrence history of this prominent crustal dislocation.

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